Expert Review of the Prototype
Note* : A video was made for the experts demonstrating how the sink is used. Questions and answers below are related to the video as well as the sink itself.

Questions	Expert 1	Expert 2	Expert 3
What is your name and your Profession/Field of Work? (Feel free to include Degrees)	John Sciotto. EVP of Preconstruction @ Genesis Engineering (BS in Civil Engineering from PSU)	Dr Joseph Sarver, Professor / Biomedical Engineering	Jaimie Dougherty, PhD. Associate Teaching Professor
If we spoke during the process of the project, what were things that you observed the team did a good job addressing, and what things did you observe we lacked, and needed more knowledge/experience in. (Optional Response)	N/A	"The team has done a good job of identifying features the user needs. The team could improve learning how to translate those features into engineering requirements (Flow rate = ## Liters/min for example)."	"The team definitely listened about decreasing complexity and considering weight. One thing that might have been helpful though was considering weight distribution, which might require more experience in physics."
Please rate this video and its helpfulness on a scale of 1 (confusing/unnecessary) to 10 (insightful/necessary)	"score of 8-9. I thought it was well done. Maybe should have dressed a little more professionally."	9	7
After watching the video, what are your thoughts on the functionality of the prototype?	"I give this prototype a solid 7-8 grade out of 10. This prototype is a solid option. I feel that the waste water	"From what I remember of the needs, the solutions	"I like it a lot. It does make me wonder how long you can run water

	disposal could be done differently. The water in the tub could get heavy for some to dispose of.”	appears to meet them all (though not sure if there is a difference between 'hot' and 'cold', or if that was necessary)”	for if the jugs are half-full as you suggest, and how heavy the whole thing is, because both elements influence the functionality. So if this were a draft I'd recommend adding in a few numbers into the video :-) like 'It can run at full water flow for x minutes' and 'if both the cold and hot waters have been emptied into the bottom it will weigh in pounds.' As is, I think you've done a great job.”
After watching the video, what are your thoughts on the prototype's ability to target student use?	“This should work for the target student use.”	“The build seems a bit unstable (weight on the back, tanks can slide around, mirror/tank assembly has a large moment arm, might break with a moderate horizontal/sideways force near the top), and I would be	“I think it will be pretty easy for students to use. My biggest concern is it accidentally getting too full for the teachers to be able to easily empty. I'd also be a little worried about it tipping - kids tend to push down on

		concerned about student use.”	things in inconvenient ways. Other than that maybe the inside of the big tub is getting gross, but it would be easy enough to wash.”
After watching the video, what are your thoughts on the longevity of the prototype as it withstands use/wear?	“I don't feel that this option has the durability to stand multiple uses. The plastic bin cover will break since it is not reinforced around the sink cut out. There are a lot of pieces. There is a bigger potential for pieces getting broken or lost.”	“I would be very concerned that the device will be damaged by an overly zealous user.”	“That is a great question. I suspect the first thing to go would be losing the pvc pipe connecting the 2 water sources - as in it getting lost. the tubing might also get damaged or kinked. If someone puts something heavy on top of the sink, will the whole thing get deformed? Those sorts of things influence longevity when you consider storage over the summer for classrooms.”

Summary of Expert Feedback:

From gathering the feedback from our external evaluation of our prototype, there were some varying opinions as well as some trends that are important to discuss. One varying opinion was that it will be either challenging for the student to use or fairly easy. Experts varied in this category, highlighting concerns of student use related to pieces not fastened in a way that could withstand damage from a young student. This further will be addressed with the team as well as the stakeholders to decide if further procedure with rebuilding pieces is necessary. One trend that was found among all the experts feedback was the prospective lack of longevity this prototype may have. All experts were concerned that the durability of the sink was lacking, and that pieces such as the PVC tubing would be damaged fairly easily. This does concern the group, as the goal in mind is for this sink to withstand 2-3 years of use without needing much replacements. Further, this concern is being addressed within the group, and may cause a proceeding into purchasing backup pieces, as well as finding new fastening solutions for pieces like the PVC pipes, to ensure that the overall functionality of the sink is not hindered by the continued use of the sink. Another issue that was seen across experts feedback was the concern of the sink's instability, or ability to tip over. While feedback from the stakeholders have not reported this issue yet, we are aware of this potential issue, and are discussing ways to prevent this from happening in the future. Overall, the experts provided key feedback that, while critical, we believe was necessary, and further allows us to review our prototype aspects, and find ways to improve and further mitigate the issues provided.